Chapter 15, Problem 30P

Problem

Find the inverse Laplace transform of:

(a) $F_1(s) = \dfrac{6s^2 + 8s + 3}{s(s^2 + 2s + 5)}$

(b) $F_2(s) = \dfrac{s^2 + 5s + 6}{(s + 1)^2(s + 4)}$

(c) $F_3(s) = \dfrac{10}{(s + 1)(s^2 + 4s + 8)}$

Step-by-step solution

Step 1 of 16

(a)

Consider the following expression of $F_1(s)$:

$$F_1(s) = \frac{(6s^2 + 8s + 3)}{s(s^2 + 2s + 5)} \quad \text{...... (1)}$$

Apply partial fracion expansion to equation 1 for simplification.

$$\begin{aligned} \frac{(6s^2 + 8s + 3)}{s(s^2 + 2s + 5)} &= \frac{A}{s} + \frac{Bs + C}{(s^2 + 2s + 5)} \\ &= \frac{A(s^2 + 2s + 5) + (Bs + C)s}{s(s^2 + 2s + 5)} \\ &= \frac{(A + B)s^2 + (2A + C)s + 5A}{s(s^2 + 2s + 5)} \end{aligned}$$

$$6s^2 + 8s + 3 = (A + B)s^2 + (2A + C)s + 5A \quad \text{...... (2)}$$

Step 2 of 16

Calculate the value of A.

Compare the coefficients constant term on both sides of this equation 2.

$$5A = 3$$

$$A = \frac{3}{5}$$

Therefore, the value of A is $\frac{3}{5}$.

Step 3 of 16

Calculate the value of C.

Comprare the coefficients of s term on both sides of the equation 2.

$$2A + C = 8$$

Substitute $\frac{3}{5}$ for A in this expression.

$$2\left(\frac{3}{5}\right) + C = 8$$

$$\begin{aligned} C &= 8 - \frac{6}{5} \\ &= \frac{34}{5} \end{aligned}$$

Therefore, the value of C is $\frac{34}{5}$.

Step 4 of 16

Calculate the value of B.

Comprare the coefficients of s^2 on both sides of equation 2.

$$A + B = 6$$
$$B = 6 - A$$

$$\begin{aligned} B &= 6 - \frac{3}{5} \\ &= \frac{27}{5} \end{aligned}$$

Therefore, the value of B is $\frac{27}{5}$.

Step 5 of 16

Write the simplified expression $F_1(s)$.

$$F_1(s)=\frac{(6s^2+8s+3)}{s(s^2+2s+5)}=\frac{A}{s}+\frac{Bs+C}{(s^2+2s+5)}$$

Substitute $\frac{3}{5}$ for A, $\frac{27}{5}$ for B, $\frac{34}{5}$ for C in this expression.

$$F_1(s)=\frac{3}{5s}+\frac{\frac{27}{5}s+\frac{34}{5}}{(s^2+2s+5)}$$
$$=\frac{3}{5s}+\frac{\frac{27}{5}s+\frac{27}{5}+\frac{7}{5}}{((s+1)^2+4)}$$
$$=\frac{3}{5s}+\frac{\frac{27}{5}(s+1)+\frac{7}{5}}{((s+1)^2+2^2)}$$
$$=\left(\frac{3}{5}\right)\frac{1}{s}+\frac{27}{5}\frac{(s+1)}{(s+1)^2+2^2}+\frac{7}{5}\frac{1}{(s+1)^2+2^2}$$
$$F_1(s)=\left(\frac{3}{5}\right)\frac{1}{s}+\frac{27}{5}\frac{(s+1)}{(s+1)^2+2^2}+\frac{7}{10}\frac{2}{(s+1)^2+2^2} \quad \ldots\ldots (3)$$

Apply inverse laplace transform to equation 3.

$$L^{-1}[F_1(s)]=L^{-1}\left[\left(\frac{3}{5}\right)\frac{1}{s}+\frac{27}{5}\frac{(s+1)}{(s+1)^2+2^2}+\frac{7}{10}\frac{2}{(s+1)^2+2^2}\right]$$

$$f_1(t)=\frac{3}{5}L^{-1}\left[\frac{1}{s}\right]+\frac{27}{5}L^{-1}\left[\frac{(s+1)}{(s+1)^2+2^2}\right]+\frac{7}{10}L^{-1}\left[\frac{2}{(s+1)^2+2^2}\right]$$
$$=\frac{3}{5}u(t)+\frac{27}{5}e^{-t}\cos 2tu(t)+\frac{7}{10}e^{-t}\sin 2tu(t)$$
$$=\left[\frac{3}{5}+\frac{27}{5}e^{-t}\cos 2t+\frac{7}{10}e^{-t}\sin 2t\right]u(t)$$

Therefore, the required inverse Laplace transform of $F_1(s)$ is

$$\boxed{\left[\frac{3}{5}+\frac{27}{5}e^{-t}\cos 2t+\frac{7}{10}e^{-t}\sin 2t\right]u(t)}.$$

Step 6 of 16

(b)

Consider the following expression of $F_2(s)$:

$$F_2(s)=\frac{s^2+5s+6}{(s+1)^2(s+4)} \quad \ldots\ldots (4)$$

Apply partial fracion to equation (4) for siplification.

$$\frac{s^2+5s+6}{(s+1)^2(s+4)}=\frac{A}{s+4}+\frac{B}{(s+1)^2}+\frac{C}{s+1} \quad \ldots\ldots (5)$$

$$\frac{s^2+5s+6}{(s+1)^2(s+4)}=\frac{A(s+1)^2+B(s+4)+C(s+1)(s+4)}{(s+1)^2(s+4)}$$
$$=\frac{(A+C)s^2+(2A+B+5C)s+(A+4B+4C)}{(s+1)^2(s+4)}$$

$$s^2+5s+6=(A+C)s^2+(2A+B+5C)s+(A+4B+4C) \quad \ldots\ldots (6)$$

Calculate the value of B.

Comprare the coefficients s^2 on both sides of this equation 6.

$$A+C=1$$

$$A=1-C \quad \cdots\cdots (7)$$

Comprare the coefficients of s term on both sides of the equation 6.

$$2A+B+5C=5$$

Substitute equation 7 in this expression.

$$2(1-C)+B+5C=5$$
$$2-2C+B+5C=5$$
$$C=\frac{1}{3}(3-B) \quad \cdots\cdots (8)$$

Comprare the coefficients of constant term on both sides of the equation 6.

$$A+4B+4C=6$$

Substitute equations 7 and 8 in this expression.

$$1-C+4B+4\frac{(3-B)}{3}=6$$
$$\frac{2}{3}(4B)-C=1$$
$$\frac{8}{3}B-\left(\frac{1}{3}(3-B)\right)=1$$
$$3B=2$$
$$B=\frac{2}{3}$$

Therefore, the value of B is $\frac{2}{3}$.

Calculate the value of C.

Substitute $\frac{2}{3}$ for B in equation 8.

$$C=\frac{1}{3}\left(3-\frac{2}{3}\right)$$
$$=\frac{7}{9}$$

Therefore, the value of C is $\frac{7}{9}$.

Calculate the value of A.

Substitute $\frac{2}{3}$ for B in equation 7.

$$A = 1 - \frac{7}{9}$$
$$= \frac{2}{9}$$

Therefore, the value of A is $\frac{2}{9}$.

Substitute $\frac{2}{9}$ for A, $\frac{7}{9}$ for C, $\frac{2}{3}$ for B in equation 5.

$$\frac{s^2+5s+6}{(s+1)^2(s+4)} = \left(\frac{2}{9}\right)\frac{1}{s+4} + \left(\frac{2}{3}\right)\frac{1}{(s+1)^2} + \left(\frac{7}{9}\right)\frac{1}{s+1} \quad \text{...... (9)}$$

Step 10 of 16

Apply inverse lalace transform to equation (9).

$$L^{-1}\left[F_2(s)\right] = L^{-1}\left[\left(\frac{2}{9}\right)\frac{1}{s+4} + \left(\frac{2}{3}\right)\frac{1}{(s+1)^2} + \left(\frac{7}{9}\right)\frac{1}{s+1}\right]$$

$$\begin{aligned} f_2(t) &= L^{-1}\left[\left(\frac{2}{9}\right)\frac{1}{s+4}\right] + L^{-1}\left[\left(\frac{2}{3}\right)\frac{1}{(s+1)^2}\right] + L^{-1}\left[\left(\frac{7}{9}\right)\frac{1}{s+1}\right] \\ &= \left(\frac{2}{9}\right)L^{-1}\left[\frac{1}{s+4}\right] + \left(\frac{2}{3}\right)L^{-1}\left[\frac{1}{(s+1)^2}\right] + \left(\frac{7}{9}\right)L^{-1}\left[\frac{1}{s+1}\right] \\ &= \frac{2}{9}e^{-4t}u(t) + \frac{2}{3}te^{-t}u(t) + \frac{7}{9}e^{-t}u(t) \\ &= \left(\frac{2}{9}e^{-4t} + \frac{2}{3}te^{-t} + \frac{7}{9}e^{-t}\right)u(t) \end{aligned}$$

Therefore, the required inverse Laplace transform of $F_2(s)$ is

$$\boxed{\left(\frac{2}{9}e^{-4t} + \frac{2}{3}te^{-t} + \frac{7}{9}e^{-t}\right)u(t)}.$$

Step 11 of 16

(c)

Consider the following expression of $F_3(s)$:

$$F_3(s) = \frac{10}{(s+1)\left(s^2+4s+8\right)} \quad \text{...... (10)}$$

Apply partial fracion expansion to equation (10) for simplification.

$$\begin{aligned} \frac{10}{(s+1)\left(s^2+4s+8\right)} &= \frac{A}{s+1} + \frac{Bs+C}{\left(s^2+4s+8\right)} \\ &= \frac{A\left(s^2+4s+8\right)+(Bs+C)(s+1)}{s\left(s^2+4s+8\right)} \\ &= \frac{(A+B)s^2+(4A+B+C)s+8A+C}{s\left(s^2+4s+8\right)} \end{aligned}$$

$$10 = (A+B)s^2 + (4A+B+C)s + 8A + C \quad \text{...... (11)}$$

Step 12 of 16

Calculate the value of A.

Comprare the coefficients constant term on both sides of this equation 11.

$$8A+C=10$$

$C=10-8A$ ...... (12)

Comprare the coefficients of s^2 on both sides of equation 11.

$$A+B=0$$

$B=-A$ ...... (13)

Comprare the coefficients of s term on both sides of the equation 11.

$$4A+B+C=0$$

Substitute equations 12 and 13 in this expression.

$$4A+(-A)+10-8A=0$$
$$A=2$$

Therefore, the value of A is 2.

Step 13 of 16

Calculate the value of B.

Substitute 2 for A in the equation 13.

$$B=-2$$

Therefore, the value of B is -2.

Step 14 of 16

Calculate the value of C.

Substitute 2 for A in the equation 12.

$$\begin{aligned}C&=10-8(2)\\&=-6\end{aligned}$$

Therefore, the value of C is -6.

Step 15 of 16

Write the simplified expression $F_3(s)$.

$$F_3(s)=\frac{10}{(s+1)(s^2+4s+8)}=\frac{A}{s+1}+\frac{Bs+C}{(s^2+4s+8)}$$

Substitute 2 for A, -2 for B, -6 for C in this expression.

$$\begin{aligned}F_3(s)&=\frac{2}{s+1}+\frac{(-2)s-6}{(s^2+4s+8)}\\&=2\frac{1}{s+1}+\frac{-2(s+2)-2}{((s+2)^2+4)}\end{aligned}$$

$F_3(s)=2\left[\frac{1}{s+1}\right]-2\left[\frac{(s+2)}{((s+2)^2+2^2)}\right]-\left[\frac{2}{((s+2)^2+2^2)}\right]$ ...... (14)

Apply inverse laplace transform to equation 14.

$$L^{-1}\left[F_3(s)\right]=L^{-1}\left[2\left[\frac{1}{s+1}\right]-2\left[\frac{(s+2)}{\left((s+2)^2+2^2\right)}\right]-\left[\frac{2}{\left((s+2)^2+2^2\right)}\right]\right]$$

$$\begin{aligned}f_3(t)&=2L^{-1}\left[\left[\frac{1}{s+1}\right]\right]-2L^{-1}\left[\frac{(s+2)}{\left((s+2)^2+2^2\right)}\right]-L^{-1}\left[\frac{2}{\left((s+2)^2+2^2\right)}\right]\\&=2e^{-t}u(t)-2e^{-2t}\cos 2tu(t)-e^{-2t}\sin 2tu(t)\\&=\left[2e^{-t}-2e^{-2t}\cos 2t-e^{-2t}\sin 2t\right]u(t)\end{aligned}$$

Therefore, the required inverse Laplace transform of $F_3(s)$ is

$\boxed{\left[2e^{-t}-2e^{-2t}\cos 2t-e^{-2t}\sin 2t\right]u(t)}$.